

All questions are compulsory.

Total Marks = 25

1. (a) In a half-subtractor circuit, the binary subtraction of two inputs of 1 bit, A and B , is performed, where: Difference (D) is the result of the subtraction ($A - B$). Borrow (Br) indicates if a borrow is needed when subtracting B from A . [1 Marks]
For example, When $A = 0$ and $B = 1$, we cannot subtract 1 from 0; so the Difference (D) = 1 and Borrow (Br) = 1. Your task is to complete the truth table given below for the half-subtractor for all combinations of inputs A and B , filling in the missing (indicated by a ?) values for the Difference (D) and Borrow (Br) for the remaining possibilities.

A	B	Difference (D)	Borrow (Br)
0	0	?	?
0	1	1	1
1	0	?	?
1	1	?	?

- (b) Similar to a full adder, in a full subtractor, there are three inputs: A , B and C . The input C could be a borrow from the previous stage. Prepare the truth table for the full subtractor based on the inputs A , B , and C , and the corresponding outputs D (Difference) and B_{out} (Borrow out). [2 Marks]
(c) Draw the circuit diagram of the full subtractor with its designated inputs and outputs. [2 Marks]
2. The following minterm function with four variables is given:

$$F(A, B, C, D) = \sum m(0, 2, 3, 4, 12, 14) + \sum d(5, 6, 7, 8, 9, 10) \quad (\text{m} = \text{minterms, d} = \text{don't care terms}) \quad (1)$$

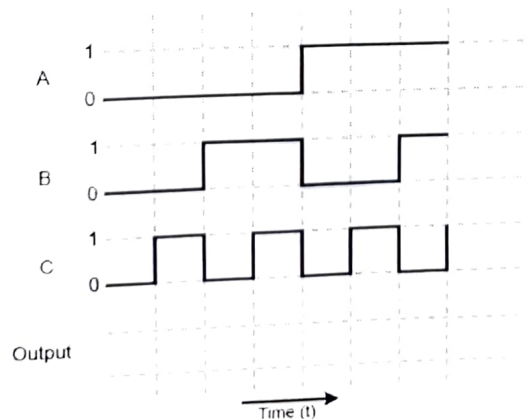
Find the simplified logical expression of the function using K-map.

[3 Marks]

3. Simplify the following expression (using axioms of Boolean Algebra, but you can use some other approach too.): [3 Marks]

$$F = ((A + (BC)')') \cdot (AB' + ABC) \quad (2)$$

4. Given the timing diagrams of 3-inputs A , B and C . Draw the output waveform of a XOR gate that takes these three variables as its inputs. [Ensure that the output is accurately derived based on the behavior of the XOR operation for the given input timing diagrams]. Assume that there is no processing delay. [3 Marks]



[Turn over

5. Write the boolean expression for the following condition: "if x is not true then y , else x ". [2 Marks]

6. You are tasked with designing a security system for a building that uses three sensors to control access. The system uses the following inputs:

- A: Motion sensor (1 if motion is detected, 0 if no motion).
- B: Door sensor (1 if the door is open, 0 if the door is closed).
- C: Security code input (1 if the correct code is entered, 0 if the code is incorrect)

The security system's alarm (Output F) will be activated ($F = 1$) if:

- Motion is detected ($A = 1$) and the door is open ($B = 1$) while the wrong security code is entered ($C = 0$).
- Motion is detected ($A = 1$) while the door is closed ($B = 0$) regardless of the security code input.

(a) Construct a truth table that represents the logic of the alarm system, showing all possible combinations of inputs (A , B , C) and the corresponding output (F). [1.5 Marks]

(b) From the truth table, derive the simplified Boolean expression for the alarm output (F). [2.5 Marks]

7. Convert the below PoS expression to its equivalent standard SoP form. [3 Marks]

$$F = (A + B + C)(A + B + C')(A + B' + C)(A' + B + C') \quad (3)$$

8. We love our 2's complement number system. We have 5 bits at our disposal. What is the largest and the smallest number that we can represent using this representation. Just state your answer, no need for explanations. [2 Marks]